

# AN ORDER-9 COUNTEREXAMPLE TO THE TRIPLICATION-TEMPLATE CONJECTURE

ABSTRACT. Ogandzhanyants, Sadov, and Kondratieva conjectured that every triPLICATION table obtained from a triPLICATION template with an admissible key is induced by a strong starter. We give a keyed template of order 9 with an admissible key such that no strong starter in  $\mathbb{Z}_{27}$  is congruous with the resulting table. A short carry identity over  $\mathbb{F}_3$  certifies nonexistence.

We use the terminology and numbering of the e-print version of [1], that is, of arXiv:2603.08381v1. For a triPLICATION table  $\Sigma_m$ , let  $\text{ST}(\Sigma_m)$  denote the set of ordered strong starters in  $\mathbb{Z}_{3m}$  congruous with  $\Sigma_m$ . Conjecture 7.1 of [1] asserts that

$$t \in K(T_0, T_1, T_2) \implies \text{ST}(\Sigma_m(T_0, T_1, T_2, t)) \neq \emptyset.$$

Unsolvable triPLICATION tables are already known. Example 7.7 of [1] exhibits two general triPLICATION tables, of orders 11 and 13, whose Modular Sudoku Problem has no solution, and the authors present them as showing that the extension of Conjecture 7.1 to all triPLICATION tables is false; Remark 7.8 there adds that such tables are rare under random sampling, reporting 27 unsolvable tables among 1000 generated tables of order 9. Neither exhibited table is asserted to arise from a triPLICATION template. What is new below is thus not an unsolvable triPLICATION table as such, but one that lies in the template class of [1, Definition 4.2] with an admissible key, that is, in exactly the class quantified over by Conjecture 7.1.

**Theorem 1.** *In  $\mathbb{Z}_9$ , let*

$$T_0 = [(1, 2), (4, 6), (5, 8), (3, 7)],$$

$$T_1 = [(4, 5), (1, 3), (8, 2), (8, 3)],$$

$$T_2 = [(5, 6), (4, 6), (7, 1), (7, 2)].$$

*Then  $\langle T_0, T_1, T_2 \rangle$  is a triPLICATION template,  $1 \in K(T_0, T_1, T_2)$ , and*

$$\text{ST}(\Sigma_9(T_0, T_1, T_2, 1)) = \emptyset.$$

*Consequently, [1, Conjecture 7.1] is false.*

*Proof.* The directed differences in each of  $T_0, T_1, T_2$ , in the displayed order, are 1, 2, 3, 4 modulo 9. The pairing  $T_0$  partitions  $\mathbb{Z}_9^*$ , while in the multiset

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union  $T_1 \uplus T_2$  every element of  $\mathbb{Z}_9^*$  occurs exactly twice. Thus  $\langle T_0, T_1, T_2 \rangle$  is a triplication template.

With key  $t = 1$ , the resulting keyed template is

$$(1) \quad \Sigma_9 = \begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & & (1, 1) & \\ 1 & (1, 2) & (5, 6) & (6, 7) \\ 2 & (4, 6) & (2, 4) & (5, 7) \\ 3 & (5, 8) & (0, 3) & (8, 2) \\ 4 & (3, 7) & (0, 4) & (8, 3). \end{array}$$

Here 0 occurs twice and every nonzero residue occurs three times. The three directed differences in regular row  $r$  are all  $r$ , and the multiset of pair sums modulo 9 is

$$\{1, 1, 1, 2, 2, 2, 3, 3, 3, 4, 4, 4, 6\}.$$

The special pair is  $(1, 1)$ , and the thirteen ordered pairs are distinct (in fact, their underlying unordered pairs are distinct). Hence (1) satisfies Definition 2.2 of [1]; equivalently,  $1 \in K(T_0, T_1, T_2)$  by Definition 4.3.

Suppose that a strong starter in  $\mathbb{Z}_{27}$  is congruous with (1). Index the thirteen table pairs by  $i$  and write the corresponding lifted pair as

$$x_i = u_i + 9U_i, \quad y_i = v_i + 9V_i, \quad U_i, V_i \in \{0, 1, 2\}.$$

Let  $r_i \in \{0, 1, 2, 3, 4\}$  be the row index, with  $r_i = 0$  for the special pair, and let  $s_i \in \{0, \dots, 8\}$  be the residue of  $u_i + v_i$  modulo 9. Define

$$\delta_i = \begin{cases} 1, & v_i - u_i < 0, \\ 0, & v_i - u_i \geq 0, \end{cases} \quad \sigma_i = \begin{cases} 1, & u_i + v_i \geq 9, \\ 0, & u_i + v_i < 9. \end{cases}$$

Then

$$(2) \quad y_i - x_i \equiv r_i + 9(V_i - U_i - \delta_i) \pmod{27},$$

$$(3) \quad x_i + y_i \equiv s_i + 9(U_i + V_i + \sigma_i) \pmod{27}.$$

All identities below are in  $\mathbb{F}_3$ .

Because the lifted pairs partition  $\mathbb{Z}_{27}^*$ , the three components above each nonzero residue modulo 9 have quotient digits 0, 1, 2; the two components above residue 0 have quotient digits 1, 2. Grouping by residue therefore gives

$$(4) \quad \sum_i (u_i U_i + v_i V_i) = 0.$$

For a fixed regular row  $r$ , put  $D_i = V_i - U_i - \delta_i$ . By (2), its three lifted directed differences are  $r + 9D_i$ . They represent distinct difference classes. Since  $r \not\equiv -r \pmod{9}$ , two of them cannot be negatives of one another; hence the three values  $D_i$  are 0, 1, 2. Summing over the regular rows yields

$$(5) \quad \sum_i r_i (V_i - U_i) = \sum_i r_i \delta_i.$$

For each  $s \in \{1, 2, 3, 4\}$ , exactly three pairs in (1) have sum  $s$  modulo 9. Their lifted sums are distinct, so by (3) the corresponding values  $U_i + V_i + \sigma_i$

are 0, 1, 2. The only other sum class is the singleton class  $s = 6$ , whose coefficient is zero in  $\mathbb{F}_3$ . Thus

$$(6) \quad \sum_i s_i(U_i + V_i) = - \sum_i s_i \sigma_i.$$

Add (4), (5), and (6). Since  $r_i \equiv v_i - u_i$  and  $s_i \equiv u_i + v_i$  modulo 9, the coefficients of  $U_i$  and  $V_i$  are respectively

$$u_i - r_i + s_i \equiv 3u_i = 0, \quad v_i + r_i + s_i \equiv 3v_i = 0.$$

Every congruous strong starter must therefore satisfy the carry identity

$$(7) \quad \sum_i r_i \delta_i = \sum_i s_i \sigma_i.$$

In (1), the only pairs with negative ordinary directed difference are (8, 2) in row 3 and (8, 3) in row 4. Hence

$$\sum_i r_i \delta_i = 3 + 4 \equiv 1 \pmod{3}.$$

The summation carries occur for

$$(5, 6), (6, 7), (4, 6), (5, 7), (5, 8), (8, 2), (3, 7), (8, 3),$$

whose sum residues are, respectively,

$$2, 4, 1, 3, 4, 1, 1, 2.$$

Their total is  $18 \equiv 0 \pmod{3}$ , contradicting (7). Thus  $\text{ST}(\Sigma_9) = \emptyset$ .  $\square$

*Remark.* Here  $T_0$  is a starter of  $\mathbb{Z}_9$ , whereas  $T_1$  and  $T_2$  are pseudostarters but not starters. Thus the example uses the pseudostarter generality of [1, Conjecture 7.1]. It leaves open the restriction to templates built entirely from starters and does not address Horton's conjecture.

#### REFERENCES

- [1] O. Ogandzhanyants, S. Sadov, and M. Kondratieva, *The triplication method for constructing strong starters*, arXiv:2603.08381v1 [math.CO] (2026); published as J. Combin. Math. Combin. Comput. **131** (2026), 115–160, doi:10.61091/jcmcc131-08.